

# PRISM: extensible sidecar metadata and validation for research data

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## Software

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## Summary

Studies of human behavior rarely produce a single kind of data: a single session can yield questionnaires, biometrics, physiological recordings, eye tracking, and brain imaging. Yet raw data files are rarely self-explanatory — a column of questionnaire responses is meaningless without knowing the item wording, each response option's label, which items are reverse-scored, or which version and language was administered.

Neuroimaging addressed an equivalent problem with the Brain Imaging Data Structure (BIDS) ([Gorgolewski et al., 2016](#)), which pairs every data file with a JSON sidecar of the same name, organizes files under predictable directory paths, and provides an official validator, making datasets portable across analysis pipelines. Its vocabulary, however, is centrally governed, meaning measurements the specification does not anticipate remain undescribed.

PRISM (Principled Research Information & Sidecar Model) — named for its commitment to the FAIR data principles ([Wilkinson & others, 2016](#)), applied at the level of the individual file rather than the repository as a whole — preserves that core principle while opening up the vocabulary. File-naming rules, modality definitions, and sidecar contracts are defined as JSON schemas that any lab can extend, so an instrument PRISM has never encountered can be fully described and validated without modifying the software itself. PRISM ships a validator, format converters, and PRISM Studio — a local graphical interface paired with a matching command-line interface. Its current modalities and template library are oriented toward psychological research, but the underlying mechanism is entirely domain-agnostic: psychology is PRISM's first application, not its limit.

## Statement of need

In practice, behavioral data is still described by a codebook: a spreadsheet or PDF appendix listing item wordings, response options, and scoring rules beside the data rather than inside it. Written for human readers, nothing enforces that they are complete, that they stay synchronized with the file they describe, or that the recorded values are among the ones they declare; and their layout is idiosyncratic to each lab, so no tool can read one reliably. Mismatches surface late — when a second analyst opens the dataset and the responses no longer match the scale they were measured on.

BIDS has expanded steadily across neuroimaging modalities, yet the community extension process through which new modalities are formally adopted can take years. Meanwhile, research data continues to outpace any central specification: new instruments, translations, short forms, and novel sensors generate measurements that remain undescribed. Capturing such data with imaging-grade rigor requires two things. First, the description contract must be open, so a lab can define a new instrument or modality itself and validate it immediately. Second, reaching a

41 valid dataset must be achievable by the researchers who collected it, not only by those who  
42 can write code: the tooling has to meet data where it already is — an Excel, CSV, SPSS, or  
43 survey-platform export with a codebook — and lead to a validated dataset through a graphical  
44 interface as readily as a script.

45 Beyond validation and portability, AI-assisted analysis, automated pipelines, and large-scale  
46 data harmonization all become more tractable when every file carries an unambiguous, schema-  
47 backed description — without waiting for a central specification to adopt it first.

48 PRISM datasets remain valid BIDS datasets, with PRISM-specific files declared in `.bidsignore`.  
49 The intended users are researchers, data stewards, and tool developers who need richer, machine-  
50 checkable descriptions than a central standard can provide — without giving up compatibility  
51 with existing BIDS tooling.

## 52 State of the field

53 Describing research data at scale requires a vocabulary that can grow where the data is produced.  
54 Any approach that enumerates measurements centrally — a specification's modality list, a  
55 curated instrument database — supports what has already been encoded and stalls on what  
56 has not, because enumeration scales with maintainer effort, not the variety of measurements  
57 researchers produce.

58 Even BIDS is not fully consistent with its own principle here: imaging files get the full data-  
59 plus-sidecar treatment, but its one home for this kind of data, `phenotype/`, falls back to a flat  
60 table, one row per participant, dropping the per-file sidecar the rest of the standard insists on.  
61 That fallback suits a score recorded once per participant, but has no place to keep session-,  
62 run-, or variant-level detail once the same instrument is administered more than once. PRISM  
63 applies the sidecar principle uniformly instead: every administration is its own sidecar-described  
64 file, with the flat `phenotype/` table available only as an optional export when matching the  
65 wider BIDS toolchain outweighs that detail.

66 The closest neighbor is Psych-DS (Kline Struhl et al., 2026), a community standard pairing  
67 CSV files under a `data/` directory with dataset-level JSON-LD metadata; it targets dataset  
68 structure and file naming rather than a per-file, per-item contract, and defines its own layout  
69 rather than remaining a valid BIDS dataset. DataLad (Halchenko et al., 2021) versions  
70 and distributes datasets without defining file contents; PRISM composes with it rather than  
71 competing, optionally recording provenance for dataset mutations and recipe scoring. Elsewhere,  
72 instrument-specific converters require a code contribution before an uncoded instrument can  
73 be described at all.

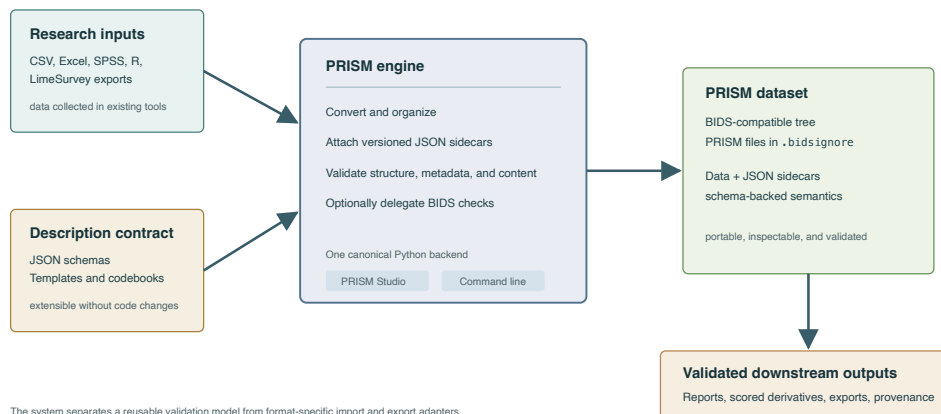
74 PRISM fixes the shape of a description and leaves its vocabulary open. An unsupported  
75 instrument becomes a data-authoring task rather than a feature request, resolvable locally and  
76 immediately without upstream approval.

## 77 Software design

78 **Implementation.** PRISM is written in Python (3.10+). The validation engine needs only  
79 `jsonschema`, `defusedxml`, and the official `bids-validator` for the optional BIDS pass, so  
80 `prism-validator` runs on a machine carrying none of the Studio stack. PRISM Studio is  
81 a local Flask application served by waitress in a native window (pywebview on macOS, a  
82 Chromium application window elsewhere), with a plain-JavaScript frontend and no build  
83 step. Converters add `pandas`, `openpyxl`, `pyreadstat`, and `pyreadr` to read and write what  
84 researchers exchange: CSV, Excel, SPSS, R, and LimeSurvey archives. Command line and  
85 interface share one core. Figure 1 shows the shared engine: adapters and contracts create and  
86 validate sidecar-described datasets.

## PRISM: a shared engine from research inputs to validated, reusable data

Declarative schemas define the contract; Studio and the command line execute the same conversion and validation workflow.



**Figure 1:** PRISM combines research data and schemas in one engine, creating and validating BIDS-compatible datasets through Studio or the command line.

87 **How validation works.** A run walks the dataset tree in four layers. *Structure*: each file path is  
88 matched against the entity grammar in `entities.schema.json` — which entities may appear,  
89 in which order, and which suffixes and extensions a modality permits — and subject and  
90 session labels are checked for consistency across the dataset. *Metadata*: every data file must  
91 have a JSON sidecar; sidecars are resolved through BIDS inheritance, merging the dataset-  
92 and subject-level files before validating the result against that modality's schema, written  
93 to the JSON Schema specification's draft-07 dialect. *Content*: for tabular modalities each  
94 value is checked against the definition its sidecar gives for that column — membership in the  
95 declared Levels, data type, numeric range — with n/a marking missing data and instrument  
96 variants honoured, so items of a long form are not demanded of a short one. *Delegation*: with  
97 `--bids`, the official BIDS Validator runs over the same tree. All findings are tagged as errors  
98 or warnings, merged into one JSON report, with a non-zero exit status on errors.

99 **Rules live in data, not code.** Entity order, modality suffixes, file extensions, and sidecar  
100 contracts are JSON schemas — that rules file plus twelve versioned schemas, six of them  
101 modalities — so adding a modality or a site-specific check means adding data rather than  
102 changing the software: a new schema, a rules entry, or — for checks no schema can express — a  
103 Python module dropped into the dataset's `validators/` directory and picked up automatically  
104 as a plugin. The cost is weaker compile-time guarantees and site-specific vocabulary, traded  
105 for extension without forking.

106 **One schema, two strictness profiles.** Each schema carries an `x-prism` block — a vendor-  
107 extension keyword, ignored by generic JSON Schema validators and read only by PRISM's own  
108 — that marks fields required only of the curated library (`officialOnlyRequired`) or only of  
109 project data (`projectOnlyRequired`), so one schema can demand curation-grade completeness  
110 of bundled templates without rejecting a researcher's in-progress questionnaire for lacking a  
111 citation.

112 **Versions coexist so validation stays reproducible.** Three schema versions (`stable`, `v0.1`, `v0.2`)  
113 ship side by side, selectable per run, so a dataset can still be validated against the rules it was  
114 authored under.

115 **Nothing is written without a preview.** `--fix` auto-repairs common issues (a missing sidecar, a  
116 missing `.bidsignore`, and the like); paired with `--dry-run` it reports what it would change

without writing anything, so a run with `--fix --dry-run` is expected before the same command without `--dry-run` applies it. Converters likewise preview what they will produce before writing, and an operation that would overwrite existing output reports the conflict instead of resolving it silently. Participant data stays on the local machine; the single network call, environmental enrichment, is opt-in.

**The current library.** The bundled library is the psychology instance of the model, not the model itself: 103 questionnaire templates and one biometrics template, each with upstream licensing recorded. A project may point PRISM at its own instead.

**Testing.** 3,730 automated tests across 268 files, plus browser-side tests for the Studio interface, run in continuous integration on every push alongside static analysis (ruff, mypy), secret scanning, and architectural-invariant checks. Data-handling paths are additionally exercised against adversarial input: a seeded generator builds a synthetic dataset of 59 deliberately hostile cases — among them session labels that must stay distinct (ses-1, ses-01, ses-pre), non-ASCII subject labels, and invalid acquisition timestamps — whose documented outcomes 51 tests assert case by case.

Usage-level material — installation, the command reference, converter walkthroughs, and the Studio guide — is maintained as documentation at <https://prism-studio.readthedocs.io> rather than reproduced here.

## Research impact statement

PRISM was built for, and used to prepare, the mixed-modality Austrian NeuroCloud dataset *Creativity: a (white) matter of connectivity* (Koschutnig, Fink, et al., 2026), whose metadata record names PRISM Studio as its creation tool; the Austrian NeuroCloud repository accepts datasets organized with PRISM. Beyond the developers' group, PRISM is used by the Department of Biological Psychology at the University of Graz and by colleagues at the University of Salzburg. It was presented to the Austrian research-data community at CLARIAH-AT's *Tag der guten Daten 2026* (Graz, February 2026), and a graduate seminar, *PRISM in Research Practice: Data Validation for Psychological Studies* (PSY.91C), runs in the winter semester 2026/27, alongside hands-on workshops at the institute. Development is public: issues raised by users outside the core team — installation failures and packaging requests — are tracked openly and addressed in subsequent releases. The repository provides tagged, archived releases (Koschutnig, Weber, et al., 2026) (currently 1.19.0), cross-platform builds, workshop materials, and example datasets including the synthetic hostile dataset, so that other groups can test the workflows locally.

## Conflict of interest

The authors declare no competing interests.

## AI usage disclosure

GPT-5.3 (OpenAI), Claude Sonnet 5 and Opus 4.6 (Anthropic), and GitHub Copilot (model/version unrecorded) assisted with code, documentation, and manuscript drafting. The authors reviewed all output, made every software-design and framing decision, and accept full responsibility for the software and this paper.

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